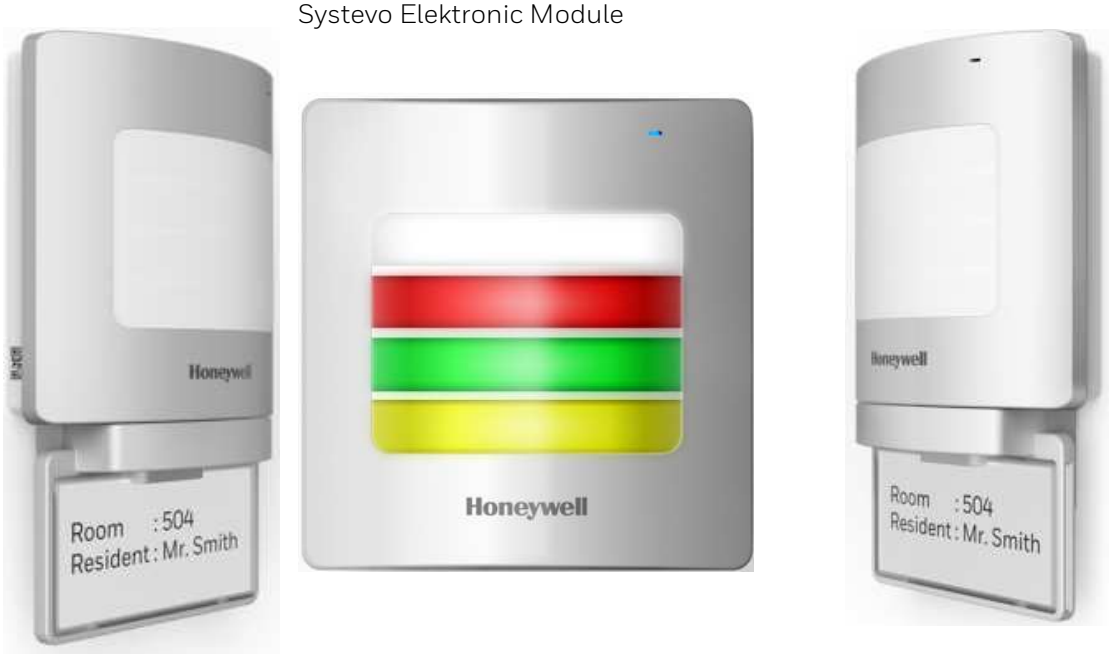
Titelseite

Systevo Elektronic Module

Systevo Elektronic Module



Systevo Electronic Module & Corridor Lamp

Installation Instruction

89470K2.GB0_V2

11.2020

GB Version

General: (TEXT wird aus LV Texte o.ä abgeleitet)

Systevo Electronic Module for pluggable light chambers with DBUS for connection of all Systevo bed units. The SEM is preconfigured ex works with standard functions. In addition, a selector switch allows meaningful operating modes of an input depending on the requirements. Thus, the SEM is equally suitable as an electronic module for patient rooms as well as function rooms. In addition, the SEM can be set via the selector switch as the central unit for the operation case call compacting set.

Equipment:

- 2 DBUS lines. Each DBUS line supports 3+1 DBUS bed units.
- 4 I / O ports preconfigured with standard functions for connecting passive components. Of these, 1 I / O port must be set via a selector switch in the mode of operation.

Safety instructions:



- In accordance with DIN VDE 0834-1, connection cables must be able to withstand a test voltage of 4 kV. The power supply of the devices can be carried in the same cable. Respect appropriate cable strength! Alternatively use system cable 89734AH / 89734AJ.
- The proper functioning of the call units is only correct if they are correctly installed. Fig. 1 ensured!
- A functional check is mandatory for safe operation in accordance with currently valid regulations (DIN VDE 0834-1: 2016-06).

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1. General / Application)

The modular concept of the Systevo electronics module is the basis for the central control unit of a room with or without speech function (SEM / SEM +) as well as a room signal light (SCL).

The Systevo electronics module (in the following SEM / SEM +) for attachable light chambers is equipped with the new data bus (DBUS) for the connection of all Systevo bed units. If a dummy cover is put on instead of the light chamber, the SEM / SEM + can be concealed without the lamp function. The SEM / SEM + is preconfigured with standard functions at the factory and is also used as a replacement for previous electronic modules in system environments from version V05.xx and in new buildings. A selector switch enables useful operating modes depending on the requirements. The SEM is equally suitable as an electronics module for patient rooms as well as for duty rooms in connection with the communication unit SCVIP. In the SEM + version, the audio bus is used for rooms with free and discrete speech. Different scopes of functions depending on the system software are considered in section 1.4 Scope of functions.

In the variant as a room signal light (SCL), the light chambers are controlled via the data bus (DBUS) in individually configurable color schemes, or each of the 4 colors via individual connections. This means that the SCL can be used in conjunction with all previous and current Systevo electronics modules, room terminals and Systevo touch terminals.

Fittings:

- 2 DBUS lines. 4 DBUS units (3 + 1) are supported per DBUS line.

- 4 I / O ports preconfigured with standard functions, for connecting passive components.

- Selector switch to determine the operating mode

- Jumper for terminating the room and audio bus.

- Lower part of the housing with connection board and lamps prepared to accommodate an attachable lamp chamber or blind cover.



• According to DIN VDE 0834-1, connecting cables including laying must withstand a test voltage of 4 KV. The power supply for the devices can be carried in the same cable. Observe the corresponding wire cross-section!

• The Systevo electronic modules SEM / SEM + and the room signal light SCL can only function properly if they are installed correctly in accordance with Chapter 2 guaranteed!

• A function check is essential for safe operation in accordance with the currently valid regulations (DIN VDE 0834-1: 2016-06).

1.1 **Requirements**

The Systevo electronics module SEM / SEM + can be used in system environments with system software ≥ V05.xx. In connection with the communication unit SCVIP, the Systevo electronic module SEM + replaces previous duty room equipment (duty room interface (DZI) with concentrated ward query (KSA). The system software from version V12.xx and, if necessary, a network connection with a data connection to the Systevo Control system center is necessary.

Range of functions

The current range of functions depending on the system software is described in the following table.

Function	System software V05 / V10 / V11	System software V12.xx
Number of lights - colors	4 (Yellow/green/red white)	4 (Yellow/green/red white)
Number of SCU on the DBUS	2	2
Max. Number of beds on the DBUS	6	3
Max. Number of beds on passive I / O terminals	2	2
Max. Number of beds in total	3	6
SBU with bed call cancelation	...	✓
Configuration of preconfigured I / O profiles	✓ Only Profile 2	✓ Profiles 1/2/3
Free call type assignment of the inputs	✓	✓
Emergency function if data communication fails	✓	✓
Power supply monitoring / measurement	✓	✓
Firmware update for SEM / SEM +	...	✓
Firmware update for components connected to the DBUS (SCL/SBU/SCU)	...	✓
Backlight of the name plate in white	✓	✓
SEM + In connection with SCVIP as a replacement for previous DZI / KSA	✓	✓

Functions, possibly subject to a license, which are only available in V12.xx systems and only at a later point in time:

- Free choice of color (multicolor) of the lights and determine individual color profiles
- Fully comprehensive input / output configuration
- Status LED for quick and easy recording of status messages (operation ok, voltage too low, emergency operation, etc.)
- Service function via buttons and LEDs for simplified installation and configuration
- Backlighting of the name badge freely selectable
- Simplified commissioning with the use of the QR code and the new commissioning app

1.2 SEM + in the duty room

The use of the SEM + as a replacement for the previous DZI service room interface enables the SCVIP communication unit to be connected to the nurse call system via the DBUS / ZBUS with a slightly reduced scope of functions. When connecting the SCVIP via the DBUS / ZBUS, it is not possible to provide the "Address / priority lists" functions. (see also the installation instructions for the SCVIP 89470J6.GB0)

2 Installation

2.1 **Mounting**

- • Carry out all assembly and installation work with the device de-energized.
- • The connection terminals are suitable for installation cables with a wire diameter of 0.6mm / 0.8mm. A wire diameter of 0.6mm is recommended for wiring in the room.
- • The selector switch for determining the operating mode is factory-set in the I- SEM (+) position.

Dimensions

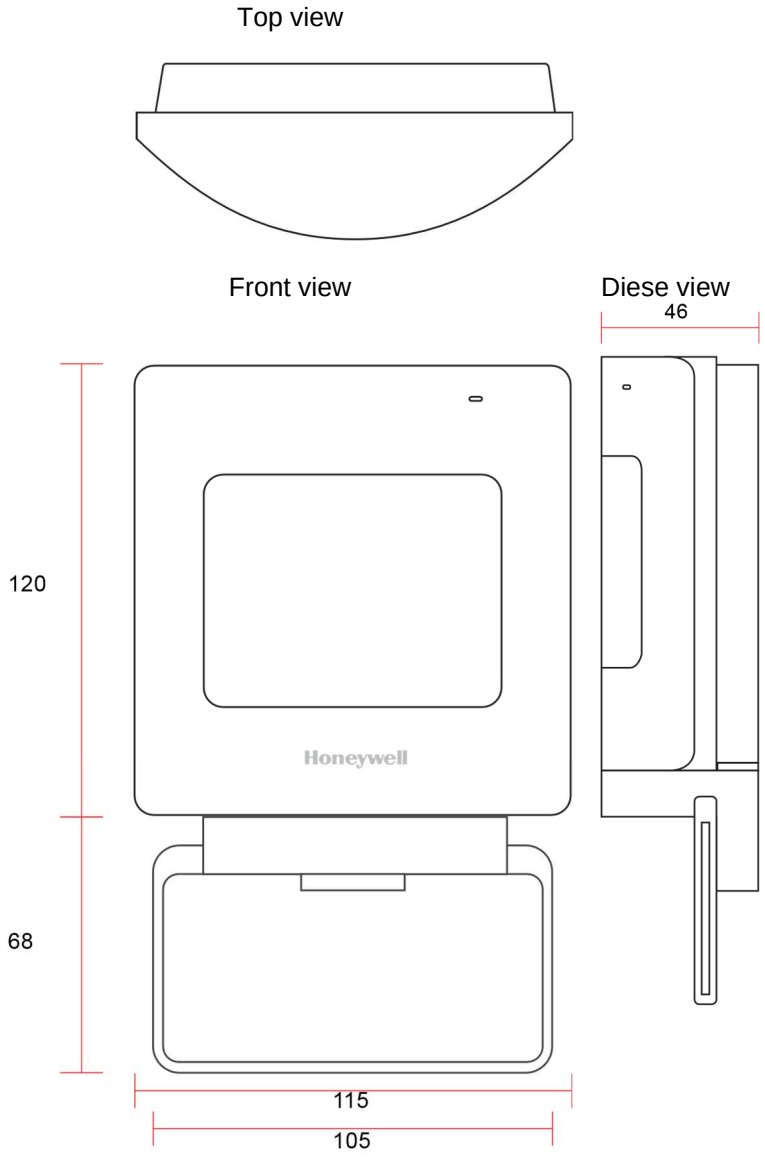


Fig. 1 Dimension Systevo Elektronic module SEM(+) with Nameplate

2.1 Remove the front lamp cover/ Open the device

The lamp cover on the front is connected to the lower part of the housing by locking lugs without screw connections and can be removed by simply pulling it off without tools.

2.2 Remove the printed circuit board from the lower part of the housing

For mounting on an installation box or on the wall, the circuit board must first be removed from the lower part of the housing. To do this, press the circuit board out of the two locking lugs on the right-hand side and remove it from the guide hooks on the left-hand side.

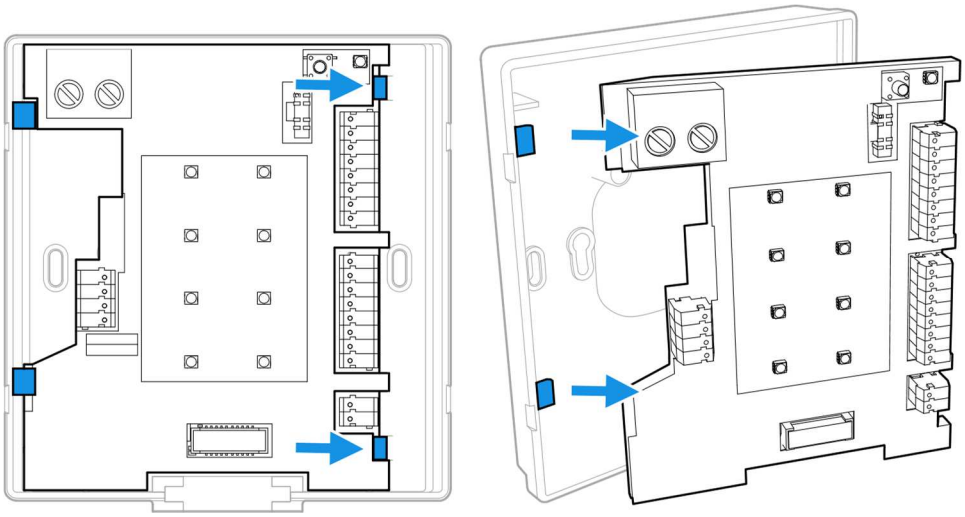
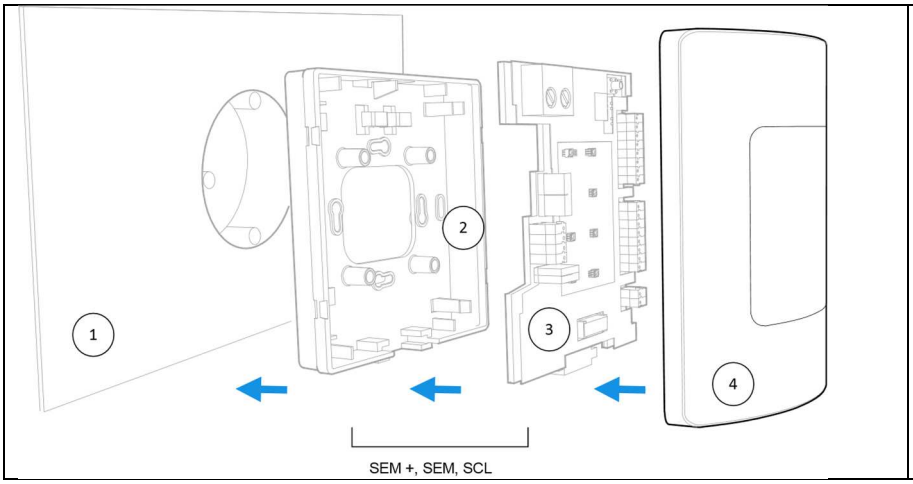


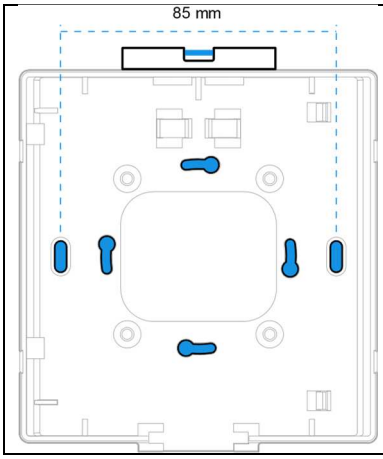
Fig. 2 Remove the printed circuit board from the lower part of the housing

2.3 Surface mounting



1	Wall
2	Lover part of the housing
3	Printed circuit
4	Front lamp cover

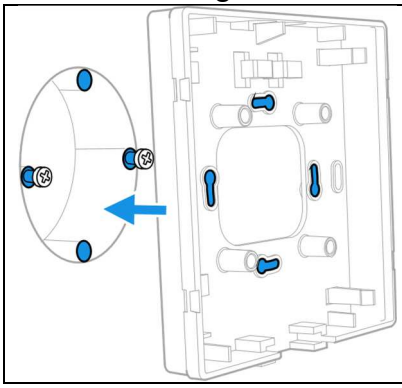
Fig. 3 Surfce Mounting



For mounting on the wall (AP), align the lower part of the housing horizontally (level) and mark the two outer mounting points on the mounting surface.
Drill two holes - according to the marked fastening points - and insert 5/6 mm dowels.
Fasten the lower part of the housing to the mounting surface with suitable screws.
Determine the choice of dowels and screws individually depending on the material of the mounting surface (cavity wall, concrete, wood, etc.).

Fig. 4 Mark the outer fastening points

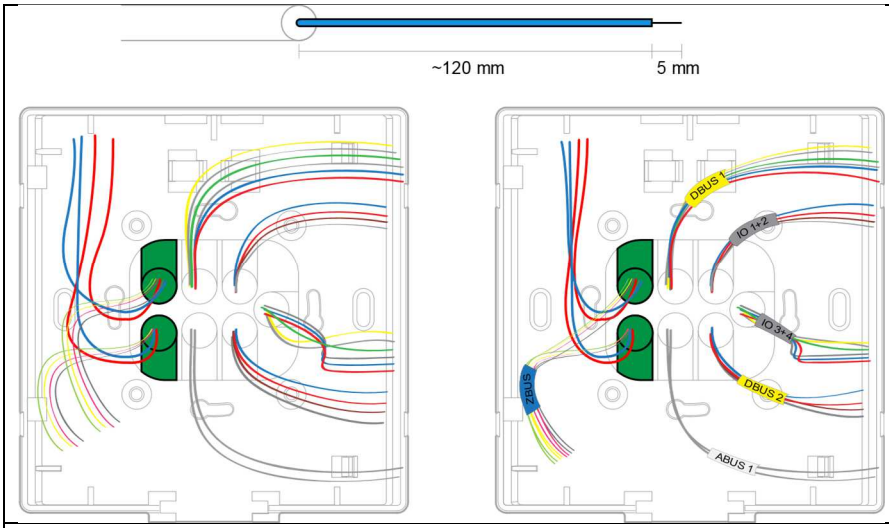
2.4 Mounting on Installation box



Place the lower part of the housing on the installation box in the correct position, align it horizontally and screw it tight with the two screws of the installation box.

Abb. 5 Mounting on a standard installation box

2.5 Cable routing when mounting on a flush-mounted installation box



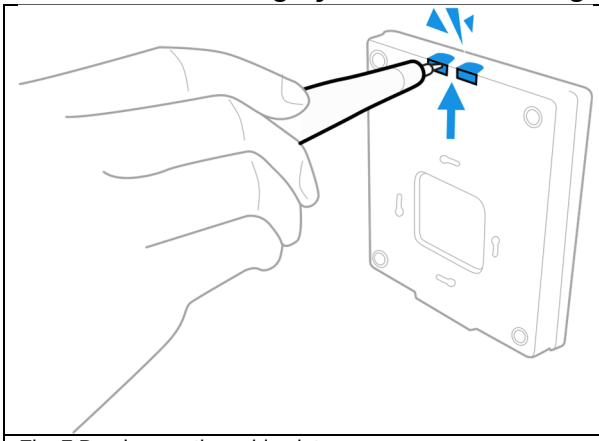
The connections on the circuit board for wiring the ward wires and the room wiring are clearly separated from each other. Ready-made labels are included in the packaging for easy identification of the installation cables.

1. 1. Strip the installation cables to approx. 120 mm and strip individual wires 5 mm
2. 2. Label the cables with the prefabricated labels
3. 3. Route the wires for the ward cabling (e.g. system cables on the left and the wires for the room cabling (DBUS, ABUS, IO) on the right

It is recommended to use cables with max. 0.6mm wires. With thicker wires, it is much more difficult to fit in them in the housing.

Fig. 6 Cable routing when mounting on a standard installation socket

2.6 Cable routing by surface mounting



Break open the cable slot on the top of the back cover.

Fig. 7 Break open the cable slot

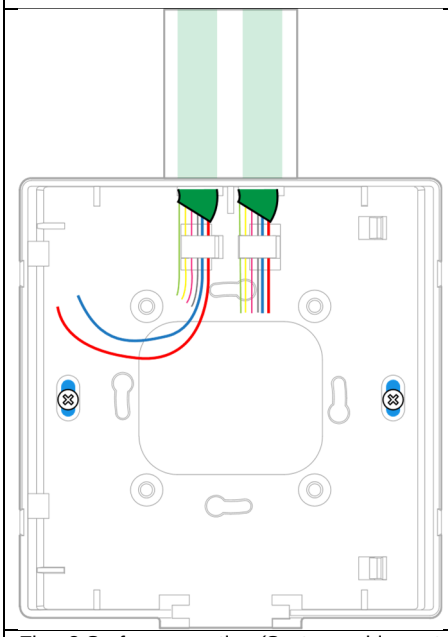
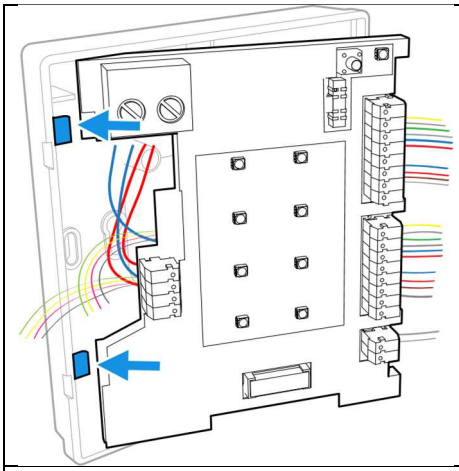
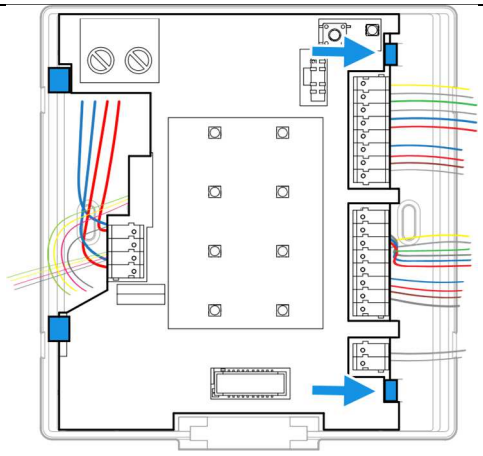


Fig-. 8 Surface mounting (System cables entering from the top of back cover)

2.7 Slide the PCB in the back cover I



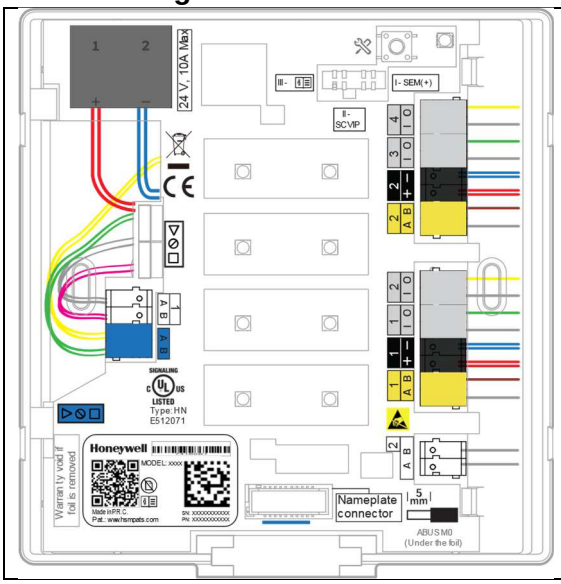
Slide the PCB inside on the left side



Push and lock it into the back cover using the snaps on the right side.

Fig. 9 Slide the PCB in the back cover

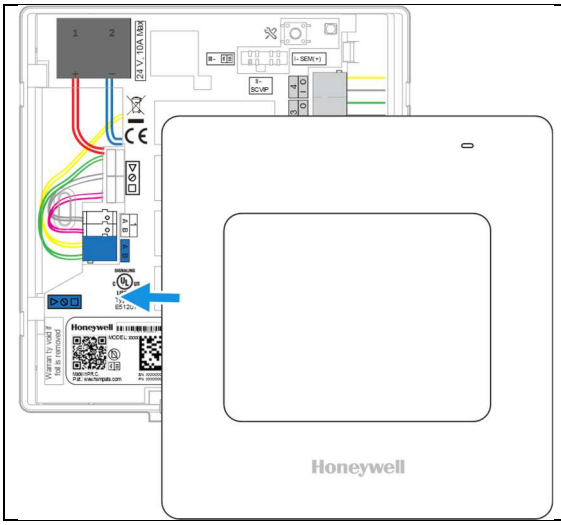
2.8 Wiring the connection cables



Attach the connection terminals and route the ward cables on the left-hand side, as well as route the cables for the room cabling, as shown in the figure on the left.

Fig. 10 Attach the connection terminals and route the cables

2.9 Lock the front cover

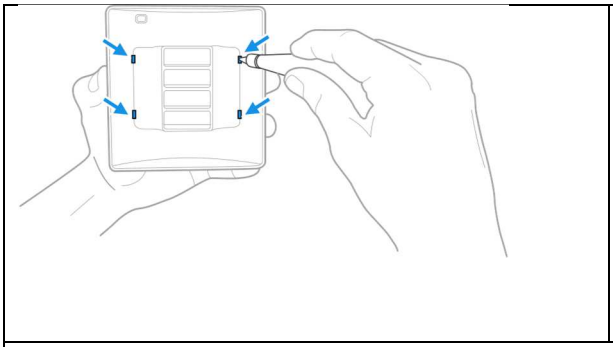


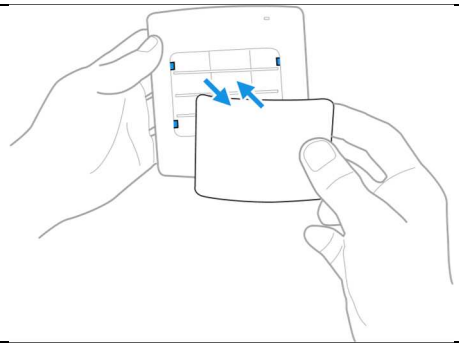
Place the cover with the light tray in the correct position and connect it to the lower part of the housing by simply pressing.

Fig. 11 Lock the front cover into the back cover, simply push and lock.

2.10 Dummy Cover

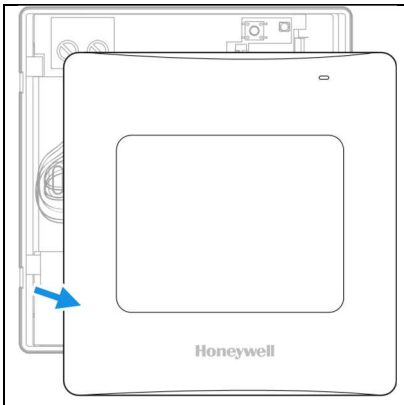
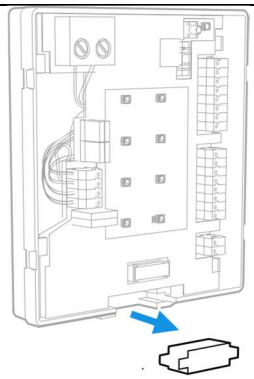
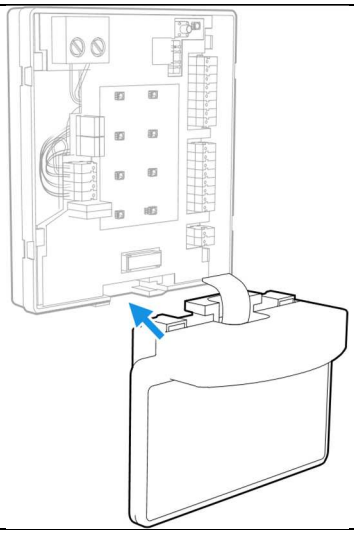
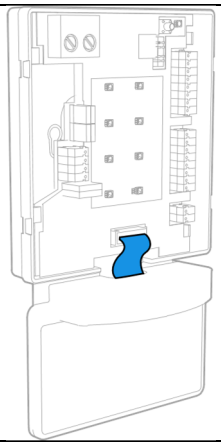
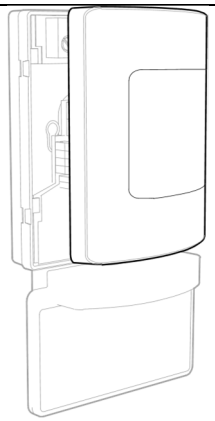
If the Systevo electronics module is concealed, the transparent insert of the cover can be exchanged for an opaque insert.



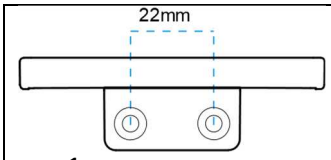


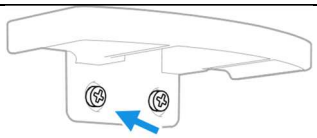
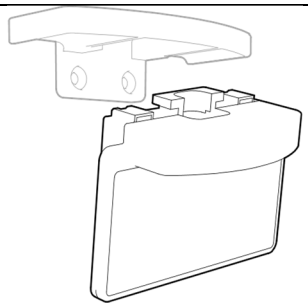
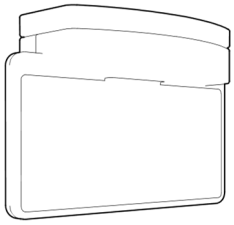
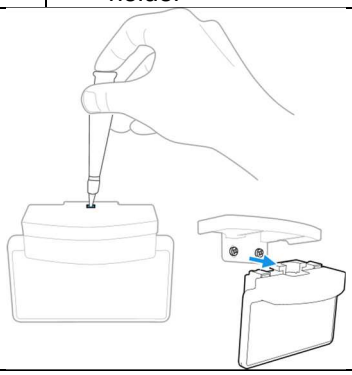
1. From the rear side of front cover, unsnap the translucent (milky) part using a tool..
2. Snap the dummy part (SKU 7259L0) into the front cover

2.11 Mounting the Nameplate at the SEM / SEM+ / SCL

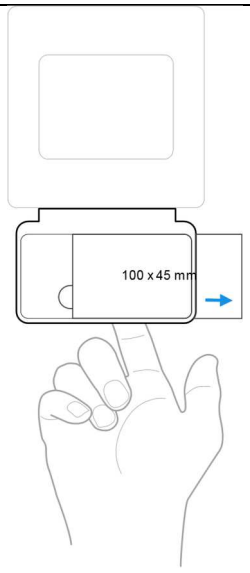
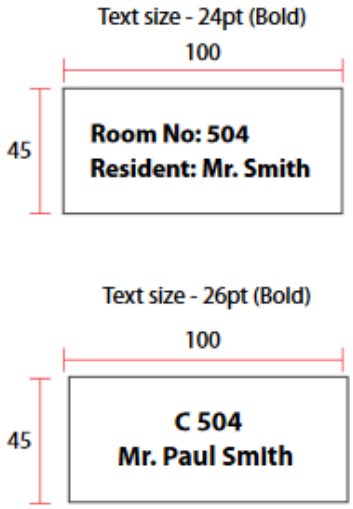
		
1. Unmount the front cover	2. Open the hole by remove the rubber part from the back cover.	3. Slide the nameplate into the back cover.
		
4. Connect the the ribbon cable to the connector on the PCBA board as shown. The blue marking on the connection cable must match the marking on the protective foil.	5. Lock the front cover.	

2.12 Nameplate standalone mounting

<table>
<tr>
<td> 

1.
Mark the drill holes with the help of the wall bracket, drill holes corresponding to the dowels (5 / 6mm) and use suitable dowels.</td>
<td> 
2.
Mount the holder on the wall</td>
<td> 
3.
Fix the nameplate on the holder</td>
</tr>
<tr>
<td>Fully assembled nameplate
  </td>
<td colspan="2">  </td>
</tr>
<tr>
<td></td>
<td colspan="2">With a tool in the slot, press lightly to release the lock and pull the name tag off the holder.
Note: Without unlocking, the mechanism can be damaged if the nameplate is removed.</td>
</tr>
</table>

2.13 Lettering of the nameplate

<table>
<tr>
<td> 
</td>
<td> 
</td>
<td></td>
</tr>
<tr>
<td>Replacing paper
Paper dimensions 100 x 45 mm Long text - Text size 24Pt Short text - Text size 26Pt A hidden opening can be font on the backside of the nameplate to slide paper in and out.</td>
<td>Recommended font sizes for the paper for labeling.</td>
<td></td>
</tr>
</table>

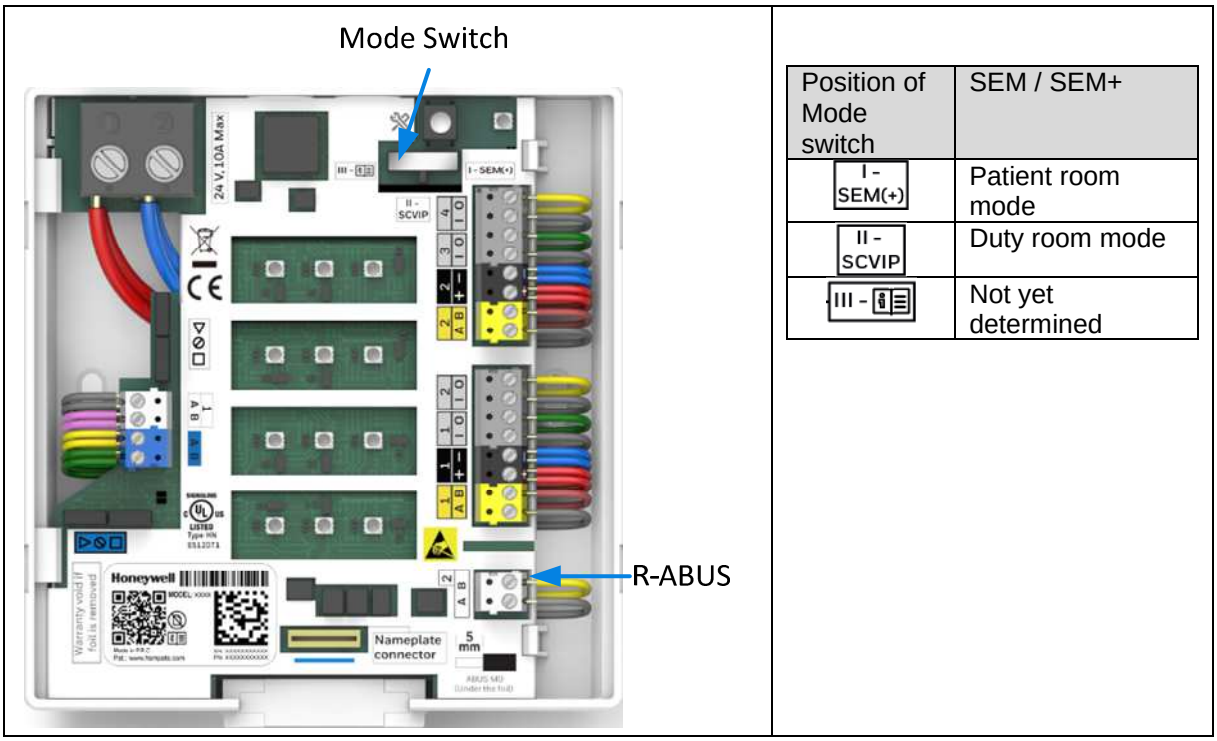
3 Electrical connection

3.1 Variants (SEM / SEM + / SCL)

The Systevo electronics module SEM / SEM + and the room signal light SCL are based on a modular design with standardized circuit boards. The different versions and the associated scope of functions can be seen in the connection terminals that are plugged in at the factory. The connection terminal marked with R-ABUS is only inserted in the SEM + version and is used to forward audio information to audio-compatible components in the room. In contrast, only the connection terminals for direct control of the light chambers and the DBUS data bus are plugged into the SCL room signal light.

3.2 Mode switch SEM / SEM+

The operating mode patient room (I-SEM +) or duty room (II-SCVIP) is set with the mode switch for the electronics module SEM +. Because of the R-ABUS and the audio functions that are possible with it, it only makes sense to connect an SCVIP communication unit to an SEM +.



Position of Mode switch	SEM / SEM+
I - SEM(+)	Patient room mode
II - SCVIP	Duty room mode
III -	Not yet determined

Fig.xy PCBA SEM / SEM+ Factory setting of Mode switch I-SEM(+)

3.3 Connection terminals on the Systevo electronics module SEM / SEM+

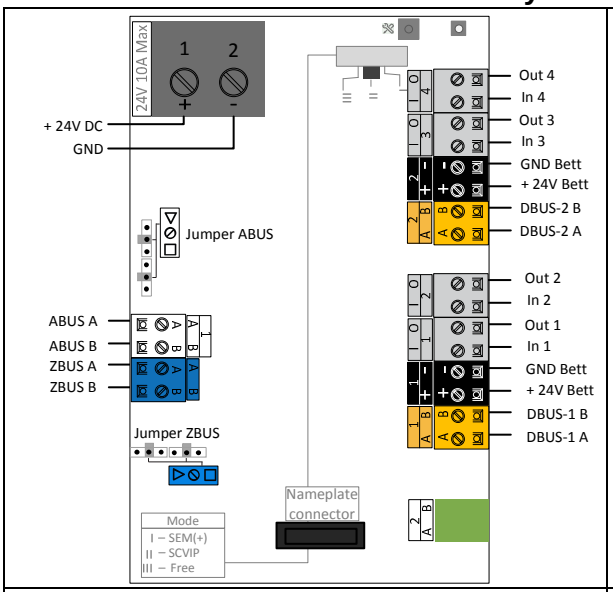


Fig.xy Terminals SEM

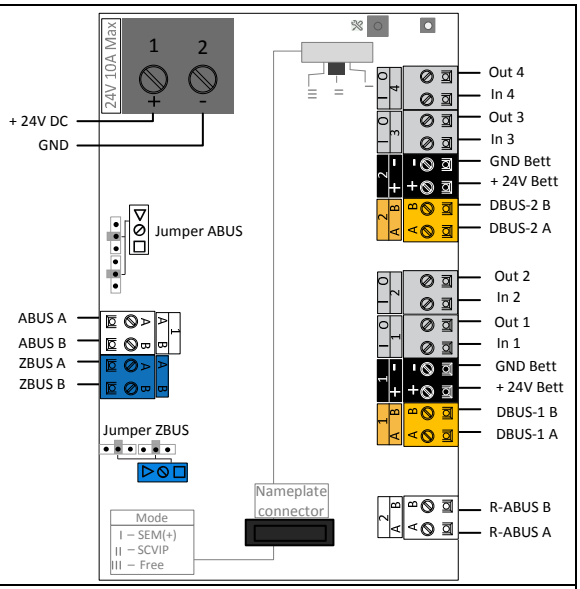
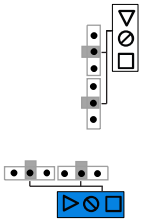
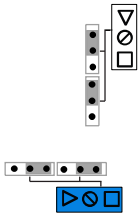
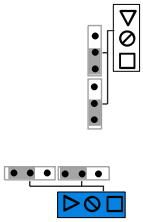


Fig.xy Terminals SEM+

3.4 Jumper Settings

The jumpers for A and ZBUS enable termination of the respective data line. In the factory setting, the termination is switched off so that the SEM / SEM + can be any participant on the ABUS / ZBUS.

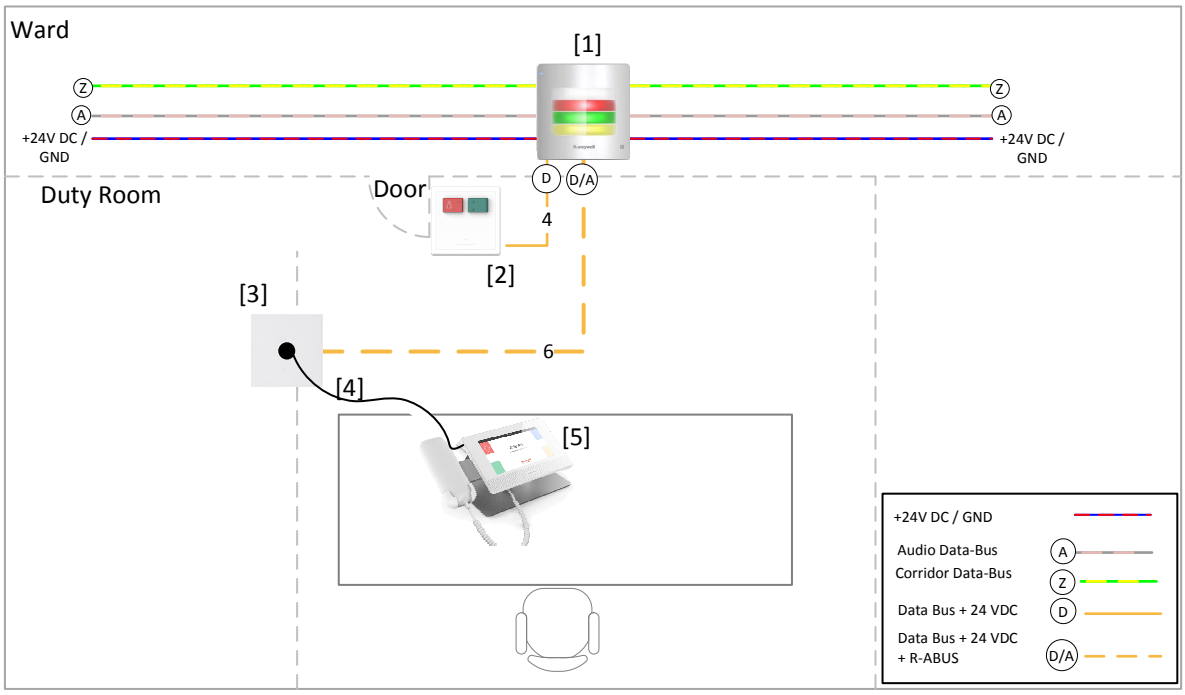
		
Factory setting: No termination -> Device is anywhere on the bus	Bus start: Device terminates the beginning of ABUS and ZBUS	Bus- End: Bus start: Device terminates the end of ABUS and ZBUS

If changes are made to the termination, these must always be made equally on ABUS and ZBUS.

3.5 SEM+ in Duty room mode

The SEM + is equipped with the room audio bus (R-ABUS) and is therefore the central control unit for a duty room with speech function for connecting the communication unit SCVIP in the SCVIP operating mode.

Schematic drawing



[1]	SEM + Systevo Elektronic Module
[2]	SCU Systevo Care Unit
[3]	SCVIP Wall unit
[4]	SCVIP Connection cable
[5]	SCVIP Systevo Care View IP

For further details, see the installation instructions for the Systevo Care View IP 89470J6.

3.6 SCL Mode switch

The operating mode and thus also the type of connection are determined with the selector switch on the SCL. The SCL enables the connection via 4-wire to components supporting DBUS and thus enables the full range of functions such as the use of predefined color profiles. In DBUS operation, up to 3 SCLs with different ID settings (ID 1 - ID3) are possible on one DBUS. In the ID2 / (WRGY) position, the four light chambers are connected directly to the light outputs of electronic modules (EM140 - EM 340), room terminals ZT99, the communication unit SCVIP and Systevo Touch Terminal STIP via 6-wire connected.

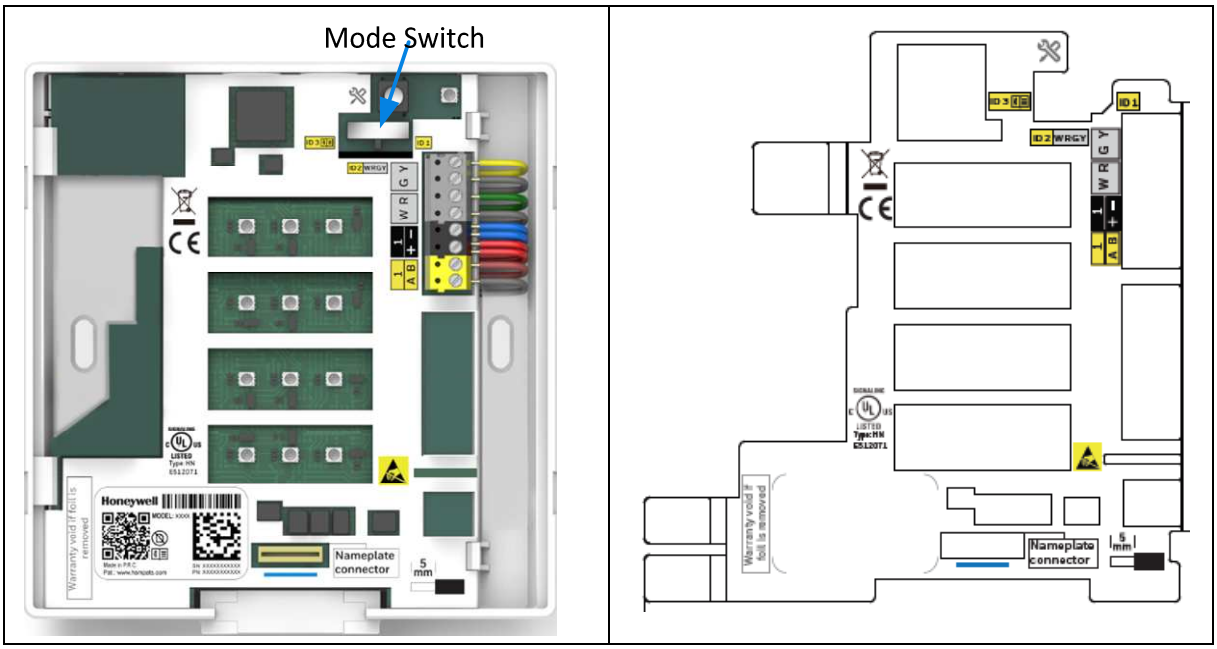
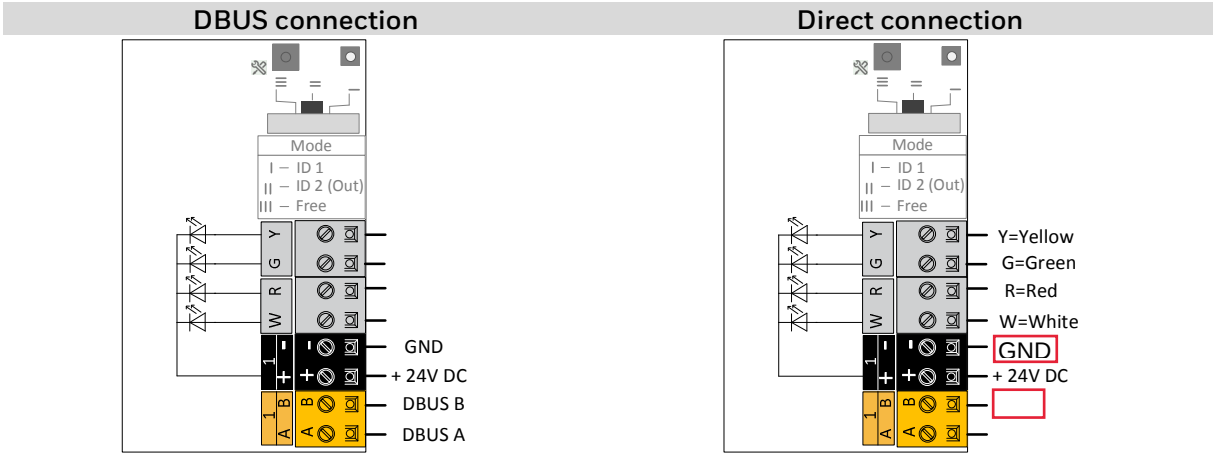


Abb.xy PCBA board SCL, Factory setting Mode switch in position ID1

Mode switch position	**DBUS** connection STIP / SCVIP	**DBUS** connection SEM / SEM+ (4 wires)	**Direct connection the chambers** (6-wires)
ID 1	1. Corridor lamp	2. Corridor lamp	Without function
ID 2 WRGY	2. Corridor lamp	3. Corridor lamp	✓
ID 3	3. Corridor lamp	4. Corridor lamp	Without function

3.7 Terminals of Corridor lamp SCL



3.8 Wiring terminals and their function:

The new Systevo wiring concept attaches great importance to simple and largely self-explanatory wiring of the components, as well as a simplified commissioning. For this purpose, the terminals are assigned color-coded functions. Any function, such as Power supply and each data bus, gets its own color for better distinction.

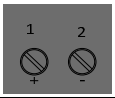
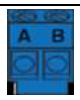
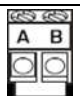
Funktion	Anschluss-klemme	Eigenschaften	SEM	SEM+	SCL
Power in 24V/10A max.		Ward Power supply, 24V max. 10A	✓	✓	...
DBUS		**DBUS:** Twisted pair, max length up to 400 m. Each DBUS host supports up to 4 DBUS devices Star-shaped wiring possible.	✓	✓	✓
ZBUS		**ZBUS:** Twisted pair, max. length 2500 m (max.500 m without repeater). The ZBUS supports up to 127 ZBUS devices (after 64 devices a repeater is necessary) Star-shaped wiring not allowed.	✓	✓	...
ABUS	 1	**ABUS:** Twisted pair, max. length 2500 m (max.500 m without repeater). The ABUS supports up to 127 ABUS devices (after 64 devices a repeater is necessary) Star-shaped wiring not allowed.	✓ Only as a support clamp	✓	...
RABUS	 2	**R-ABUS: R-Audio Bus** Twisted pair, max length up to 200 m. Each DBUS host supports up to 2 DBUS devices	...	✓	...
4 x NC (up) / passive I/O down	 1-4	**I/O Port:** NC units can be connected I-> Call circuit in 0V~24V +/- 10% DC. O-> Output for reassurance lamp max. 100mA (open Collector zo GND)	✓	✓	...
2 x 2 Input	 WR GY	Input Terminals for the direct 4 color chamber connection	...	...	✓
2 x Power for room modules	 1-2	**Power supply for beds**, max. 2A (all power supply terminals for BBUS + DBUS 1 + 2 together) * 1 : Terminator 1 for power supply for SCL	✓	✓	✓*
✓ = available		... = unavailable			

Fig.1 Table of terminals and related functions.

3.9 Factory setting standard functions SEM / SEM+

Fixed standard functions and connection terminals assigned to the functions significantly simplify installation and commissioning and often make complex configuration superfluous. Pre-assigned standard functions are retained even if the data communication with the Systevo Control fails or is disrupted. For this purpose, the Systevo electronics modules are preconfigured ex works with a defined profile (Profile2) and can therefore also be used in system environments V05.xx / V10.xx / V11.xx as a replacement for previous electronics modules. Profiles 1 + 3 can only be selected in system environments from V12.xx onwards.

Profile 1:
This configuration type stands for applications **without presence 2**, for **1 bed** and the connection of an additional passive corridor lamp with **3 chambers** (white, red and green).

Features:

- Bath / WC call and cancel button WC
- 2 SCU for Bath / Door (Room & Bed Call cancellation)
- Detection of plug shedding with identification
- Reassurance light function is controlled by the connection unit. This means that a locating light function is also available on plugged-in call units that are operated on passive ejection units (74199 and 74189x).
- plug out call
- **No** presence 2

I/O Port		Description of the function
1	I	Input for Bad/WC call (RTWC)
	O	Output for Bad/WC lamp (RLWC)
2	I	Input for cancel button Bad/WC (ATWC)
	O	Output for presence lamp (AL1)
3	I	Do not use, No Function
	O	Output for call (RL)
4	I	Input for Bed 1 call (RT)
	O	Output for reassurance lamp for Bed call (BL)

Profile 2:

This configuration type stands for applications with up to two beds and the connection of an additional passive corridor lamp with 4 chambers (white, red, green and yellow).

Features:

- Bed call with identification
- 2 SCU for Bath / Door (Room & Bed Call cancellation)
- Bath / toilet call and cancel button
- Detection of plug out (without identification and not on auto release plug units 74199 and 74189x)
- Local location light (not on auto release plug units 74199 and 74189x)
- Combined lamp output for all bed and room calls (no differentiation)
- Plug out call and find light function only on call units (73075xx) and bed units (SBU-Dxx and SBU-Mx).

I/O Port		Description of the function
1	I	Input for Bad/WC call (RTWC)
	O	Output for Bad/WC lamp(RLWC)
2	I	Input for cancel button Bad/WC (ATWC)
	O	Output for presence lamp (AL1)
3	I	Input for Bed 2 call (RT Bed 2)
	O	Output for lamps for all bed - and patient room calls (RL+BL) (Without bed identification)
4	I	Input for Bed 1 call (RT)
	O	Output forLpresence 2 lamp (AL2)

Profile 3:

This configuration type stands for applications **with presence 2**, **DIA-call**, with up to **three beds** on the DBUS and the connection of an additional passive corridor lamp with **4 chambers** (white, red, green and yellow).

Functions:

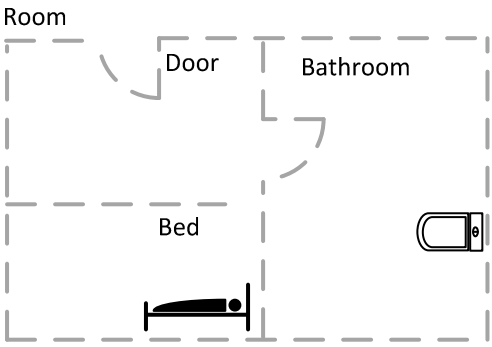
- 3 Components SBU-xx at the DBUS
- 2 SCU for Bath / Door (Room & Bed Call cancellation)
- Bedside call with identification
- Bedside call canceling
- Bedside Diagnostic call
- Diagnostic call for room
- Plug release detection
- Local finding light
- Bathroom/WC call and canceler

I/O Port		Description of the function
1	I	Input for Bad/WC call (RTWC)
	O	Output for Bad/WC lamp(RLWC)
2	I	Input for cancel button Bad/WC (ATWC)
	O	Output for presence lamp (AL1)
3	I	Input for Diagnostic call for room (DIA)
	O	Output for all room calls (RL)
4	I	Input for presence 2 (AW2)
	O	Output forLpresence 2 lamp (AL2)

4. Room variants

4.1 Room assembly in connection with a SEM / SEM+

A room can be composed of three different areas. A variant can be determined for each sub-area (door, bathroom / toilet and bed).



Room = R
Door = D
Toilet = T
Point of Care = P

R= D + T + P + P

4.2 Door variants (D)

Door	Figure	Basic Features	Free speach	SEM	SEM+	DBUS 1	DBUS 2
D1		PR1 / Call and buzzer	...	✓	✓	1	...
D2		SRD Display with PR1 / Call and buzzer	...	✓	✓	1	...
D3		SRT Display with PR1 / Call and buzzer + Audio	✓	...	✓	1	...

4.3 Bathroom / Toilet variants (T)

Bathroom / WC	Figure	Feature	4-Wire (DBUS)	5-Wire (IO-Port)	DBUS 1	DBUS 2
T1		Cancel button + max. 5 call button with pull cord at passive IO	...	✓	...	...
T2		Call / Cancel button + max. 4 call button with pull cord at passive IO	...	✓	...	...
T3		Call / Cancel button + max. 4 call button with pull cord at DBUS	✓	...	1	...

4.4 PoC (P) variants:

PoC	Figure	Feature	Free Speach	Discrete Speach	SEM	SEM +	DBUS 1	DBUS 2
P1		Call button + 7 pol. plug	...	...	✓	✓	max. 3	max. 3
P2		Call button + 7 pol. plug + magnetic plug	...	...	✓	✓	max. 3	max. 3
P3		Call button + 7 pol. plug + magnetic plug	✓	...	...	✓	1	max. 2
P4*		Call button + 7 pol. plug + magnetic plug	✓	✓	...	✓	1	max. 2
P5		Magnetic plug 2/5 domotic output	...	...	✓	✓	max. 3	max. 3

P4 * - A Systevo Audio Switch (SAS) must be provided for each DBUS to switch between TV sound and discrete speech.

4.5 Combinations in the patient room

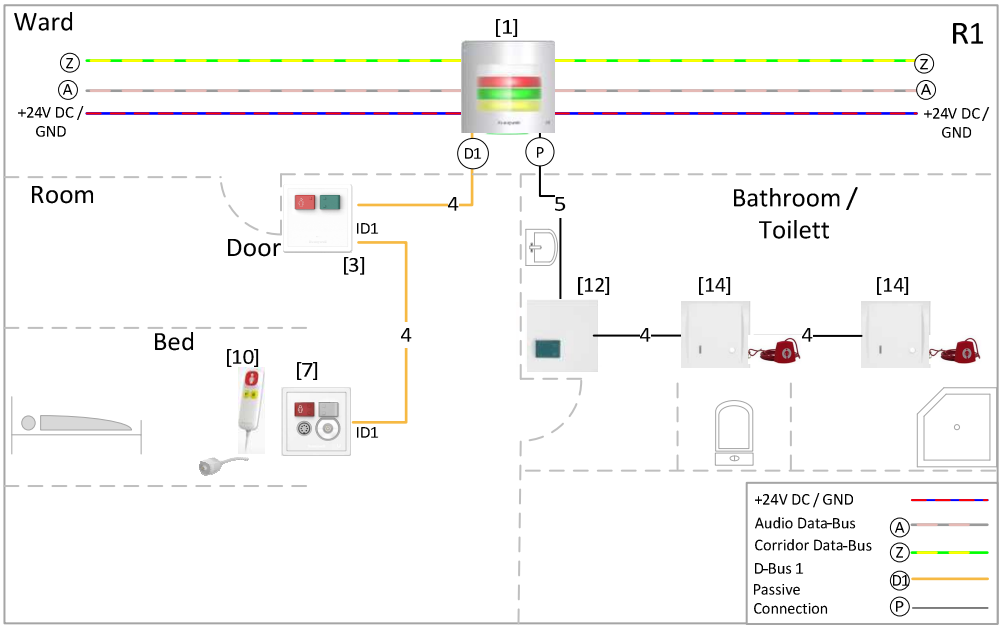
The table below shows possible room combinations compiled from the above variants. In compliance with the restrictions for operation with the DBUS (max. 4 DBUS components, of which max. 3 are of the same type plus another DBUS component of a different type on a DBUS line), the rooms are separated from the individual areas of door, bed and bathroom / WC assembled.

Accordingly, the following applies to a room: R= D+P(+P)+T

Type	Description	Amount of DBUS	**D**oor	PoC	Visitors	Bathroom / WC (**T**oilett)
R1	1 Bed patient room with SBU-DM2+; Bathroom / WC with cancel button ATWC	2	D1	P2	...	T1
R2	2 Bed patient room with SBU-DM2+; bathroom / WC with call-/ cancel button	3	D1	2 x P2	...	T2
R3	3 Bed patient room with STM with SBU-DM2+ & visitors' area, with SBU-D2+; Bathroom / WC with cancel button ATWC	5	D2	3 x P2	P1	T1
R4	6 Bed patient room with SBU-D2 & 4 wire cabled bathroom / WC with call-/ cancel button	8	D1	6 x P1	...	T3
R5	3 Bed patient room with door side free speech and bed side discrete speech 4 wire cabled bathroom / WC with call-/ cancel button & visitors' area	6	D3	3 x P4	P2	T3
R6	6 Bed patient room with door side free speach & 4 wire cabled bathroom / WC with call-/ cancel button	8	D3	6 x P3	...	T3
R7	4 Bed patient room with door side free speech and bed side discrete speech & bed side TV-Ton & 4 wire cabled bathroom / WC with cancel button ATWC (an audio switch SAS required for each DBUS)	6+2	D3	4 x P4	...	T3

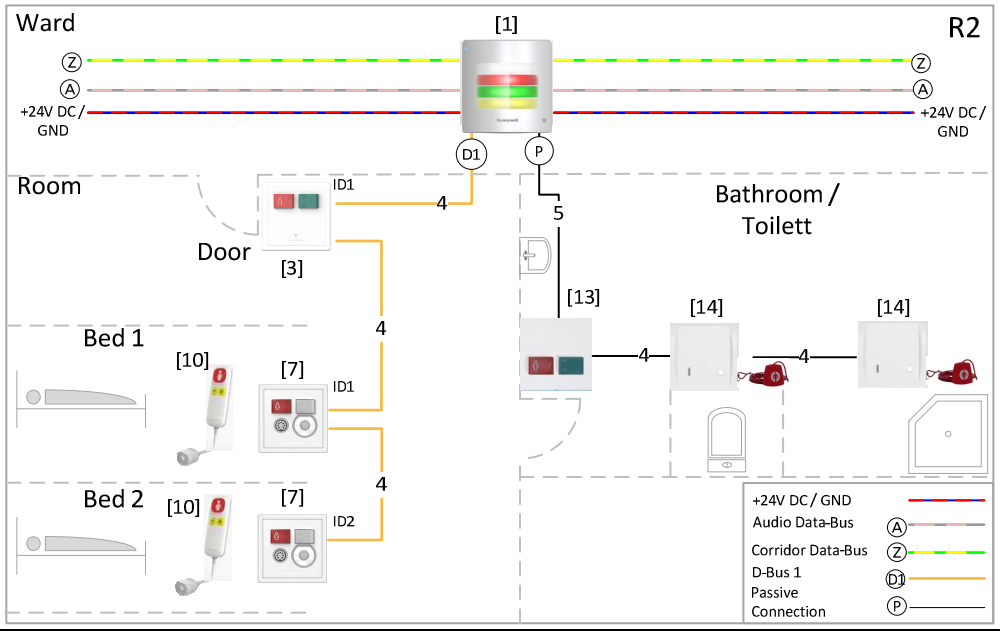
4.5.1 1 Bed patient room with SBU-DM2+; Bathroom / WC with cancel button ATWC

R1	1 Bed patient room with SBU-DM2+; Bathroom / WC with cancel button ATWC



4.5.2 2 Bed patient room with SBU-DM2+; bathroom / WC with call-/ cancel button

R2	2 Bed patient room with SBU-DM2+; bathroom / WC with call-/ cancel button

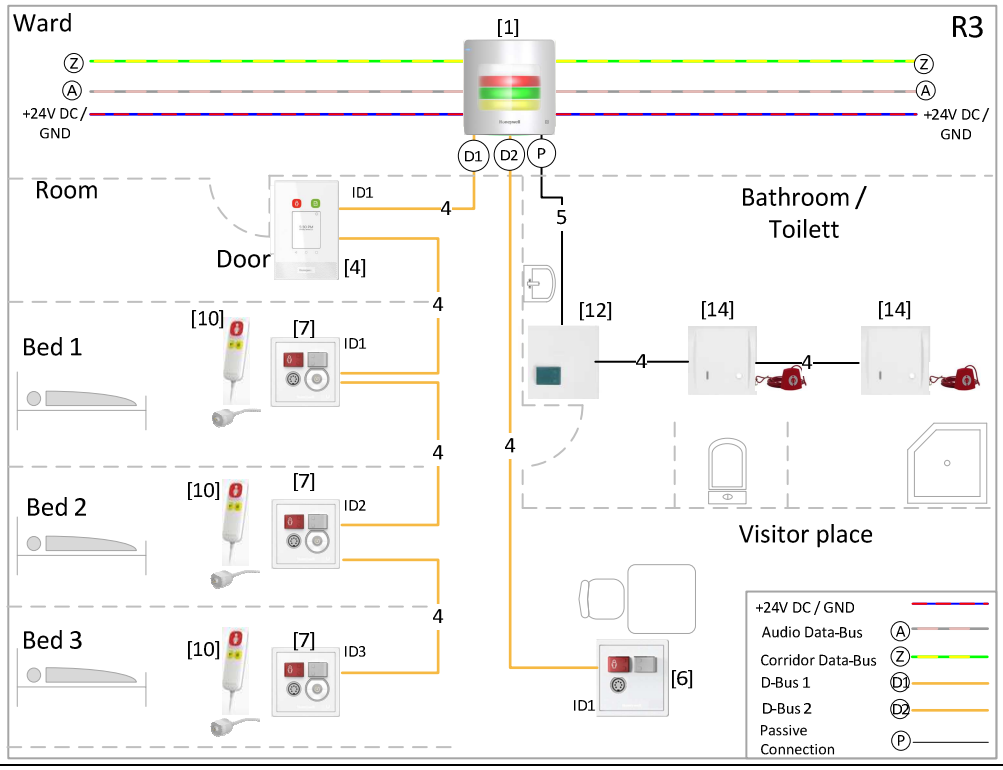


[1]	SEM
[3]	SCU
[7]	SBU-DM2+
[10]	Systevo Call Easy Magnet
[12]	73642C
[13]	73642B3
[14]	70045A3 / 70006Bx

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

4.5.3 3 Bed patient room with SRD with SBU-DM2+ & visitors' area, with SBU-D2+; Bathroom / WC with cancel button ATWC

R3	3 Bed patient room with SRD with SBU-DM2+ & visitors' area, with SBU-D2+; Bathroom / WC with cancel button ATWC

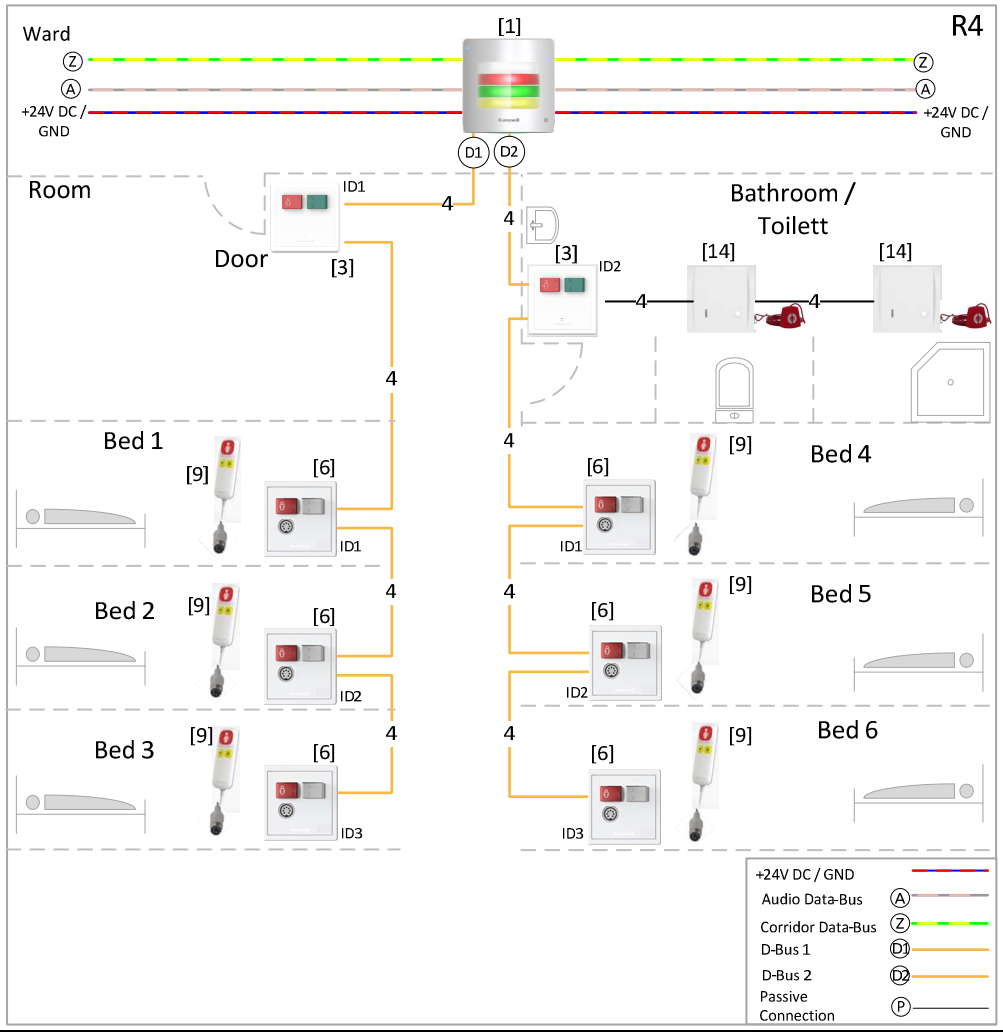


[1]	SEM
[4]	SRD
[6]	SBU-D2+
[7]	SBU-DM2+
[10]	Systevo Call Easy Magnet
[12]	73642C
[14]	70045A3 / 70006Bx

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

4.5.4 6 Bed patient room with SBU-D2 & 4 wire cabled bathroom / WC with call-/ cancel button

R4	6 Bed patient room with SBU-D2 & 4 wire cabled bathroom / WC with call-/ cancel button

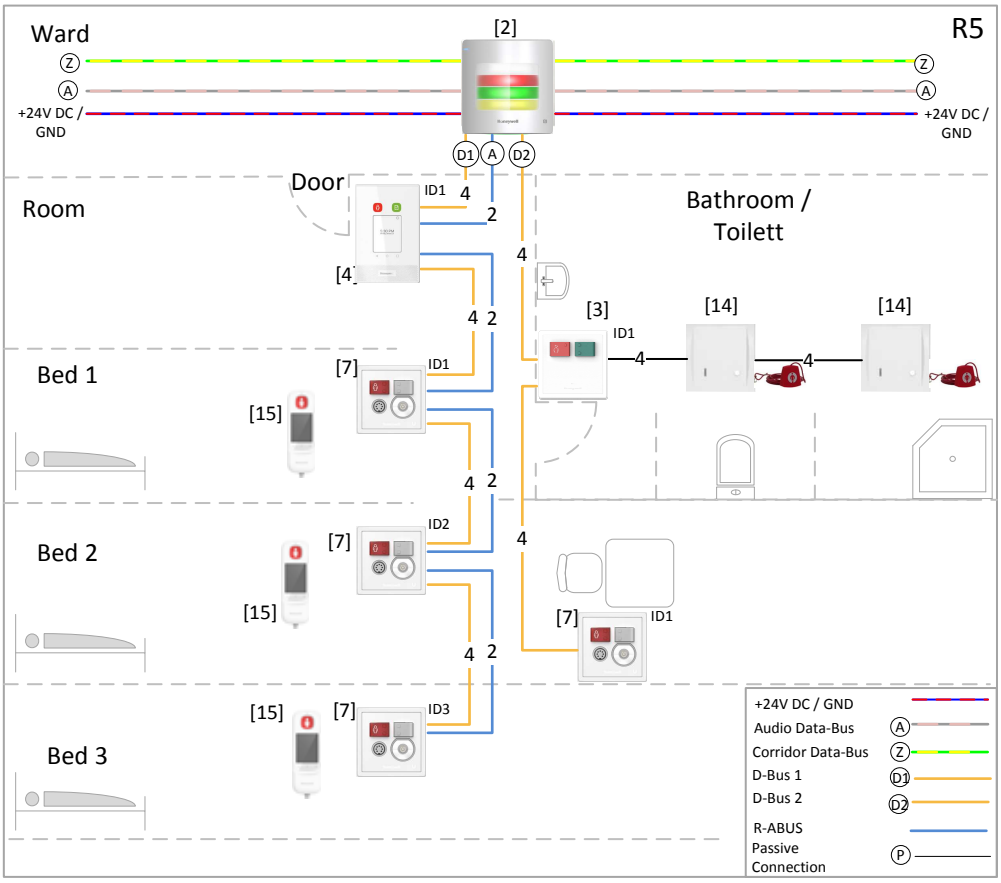


[1]	SEM
[3]	SCU
[6]	SBU-D2+
[9]	Systevo Call Easy 7 pol
[14]	70045A3 / 70006Bx

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

4.5.5 3 Bed patient room with door side free speech and bed side discrete speech, 4 wire cabled bathroom / WC with call-/ cancel button & visitors' area

R5	3 Bed patient room with door side free speech and bed side discrete speech 4 wire cabled bathroom / WC with call-/ cancel button & visitors' area

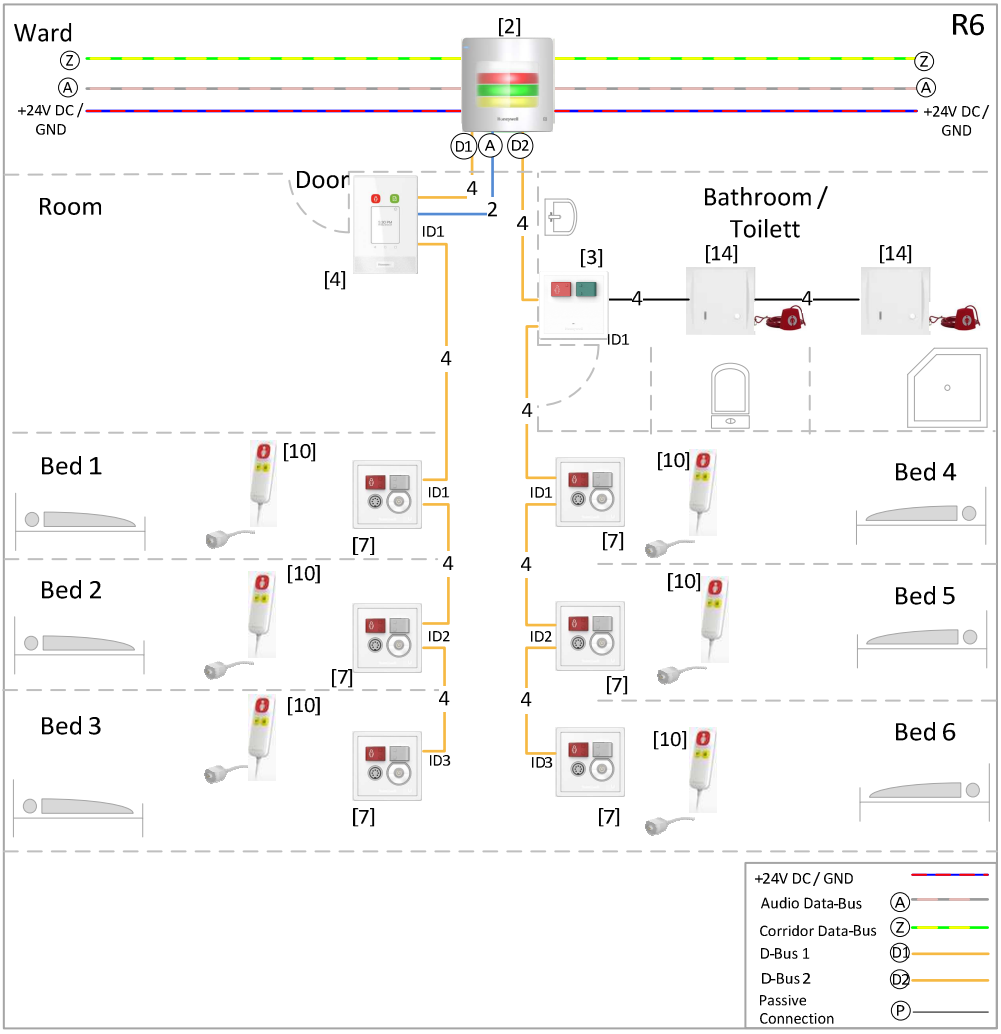


[2]	SEM+
[3]	SCU
[4]	SRT
[7]	SBU-DM2+
[14]	70045A3 / 70006Bx
[15]	Systevo Com IP

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

4.5.6 6 Bed patient room with door side free speach & 4 wire cabled bathroom / WC with call-/ cancel button

R6	6 Bed patient room with door side free speach & 4 wire cabled bathroom / WC with call-/ cancel button

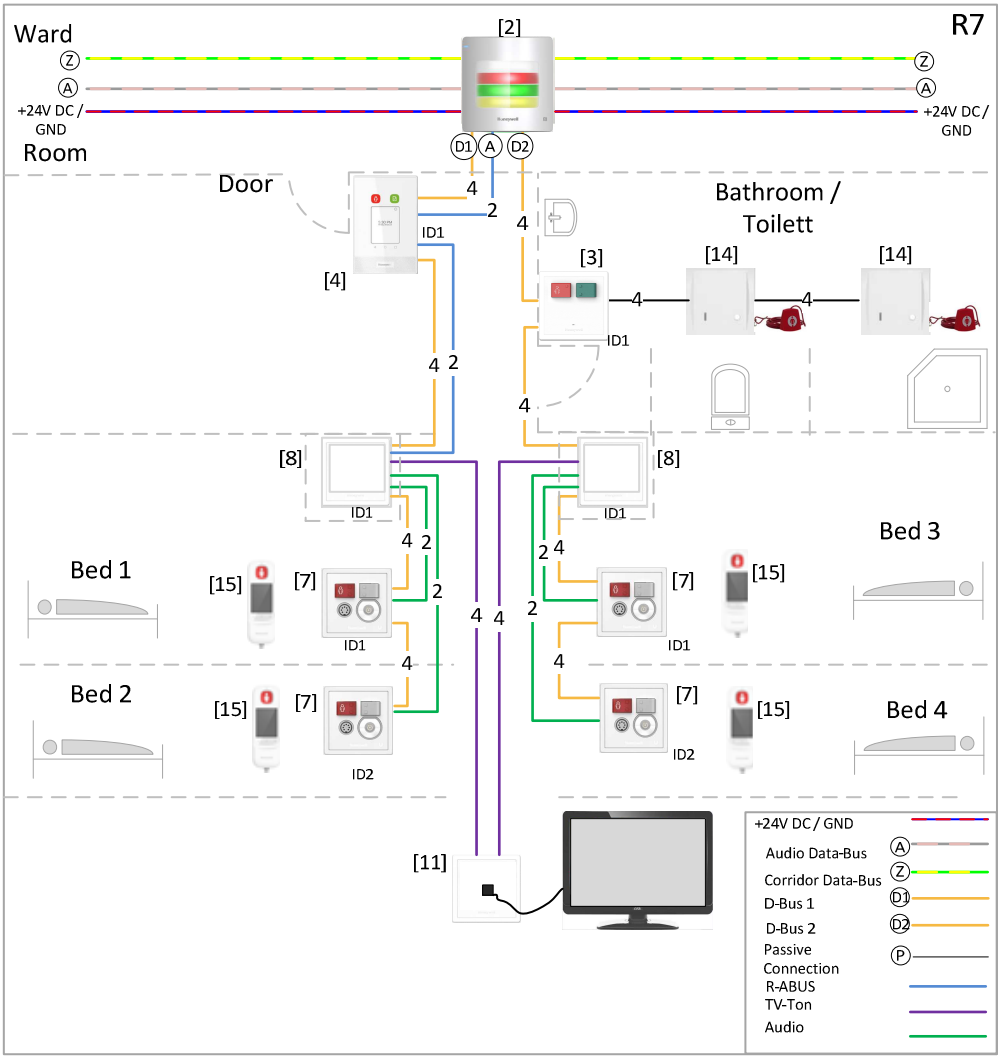


[2]	SEM+
[3]	SCU
[4]	SRT
[7]	SBU-DM2+
[10]	Systevo Call Easy
[14]	70045A3 / 70006Bx

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

4.5.7 4 Bed patient room with door side free speech and bed side discrete speech & bed side TV-Ton & 4 wire cabled bathroom / WC with cancel button ATWC (an audio switch SAS required for each DBUS)

R7	4 Bed patient room with door side free speech and bed side discrete speech & bed side TV-Ton & 4 wire cabled bathroom / WC with cancel button ATWC (an audio switch SAS required for each DBUS)



[2]	SEM+
[3]	SCU
[4]	SRT
[7]	SBU-DM2+
[8]	SAS
[11]	SAI
[14]	70045A3 / 70006Bx
[15]	Systevo Com IP

Please note: at the time of publication, the components shown may not yet be available. Be sure to check availability.

5. Notes on commissioning / configuration

Commissioning

5.2 SEM / SEM + in V5.xx / V10.xx / V11.xx environments

The data structure of the Systevo electronics module SEM / SEM + is based on the properties and functionality of the room terminal ZT99 and is comparable. When replacing a Systevo electronic module of the type EM 140/230/340 as well as a DZI, it should be noted that the automatic integration is not possible, but via the manual way of ZBUS addressing (Service 65) and subsequent creation of the database (Service 63). An SEM / SEM + integrated in this way will appear in the configuration program cp99config as the room terminal ZT99.

Procedure when integrating a SEM / SEM +:

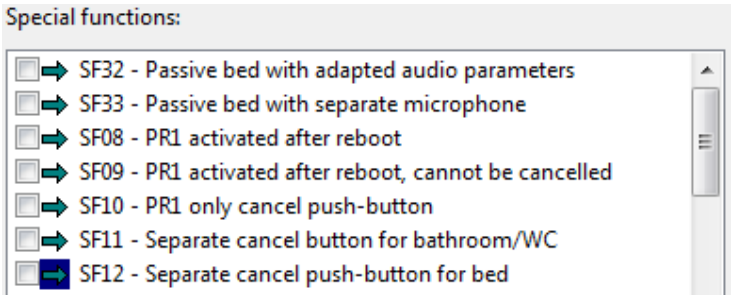
- Record all properties of the electronic module to be exchanged, such as:
- Room name, nurse call group
- Input-output type
- special functions
- Manual integration of the SEM / SEM + through ZBUS addressing -> With TerraTerm command: S65
- The SEM / SEM + is represented as ZT99
- Manual creation of the database -> With TerraTerm command: S63
- Configuration of the properties, possibly creating and assigning a comparable input / output profile for room terminals ZT99.
- Save the configuration and transfer it to the system.
- Perform a functional test.

Configuration

General:

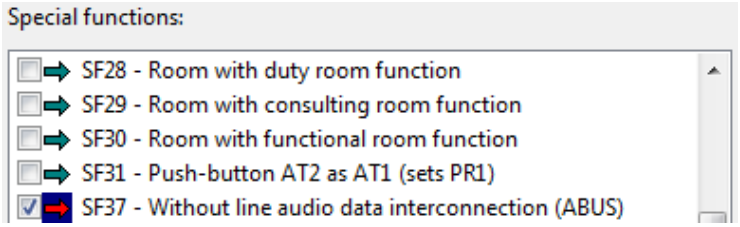
1.2 Separate cancel push-button for bed

In the case of mixed operation of bed units SBU-M2 / M5 with bed units SBU D2 / DM2 +, the special parameter **SF12** "Separate cancel push-button for bed" must **not** be set.



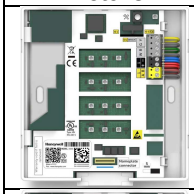
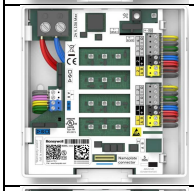
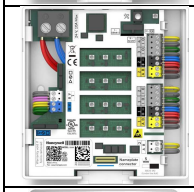
Only SEM+:

If no audio-capable components are connected to the R-ABUS of the SEM +, the check mark must be set when specifying the special functions for SF37 “Without line audio data interconnection (ABUS).



6. Scope of delivery / accessories

The Systevo electronics module SEM / SEM + and the room signal light SCL are available as a housing base with the connection board in white and are provided with covers in white and silver. An opaque insert for the cover is available for concealed installation where the lights are not required. The name tag is also available in white and silver to match.

Picture	SKU	Title	Short name
	72592D	Systevo Corridor Lamp	SCL
	72590D	Systevo Electronic Module	SEM
	72591D	Systevo Electronic Module plus with Audio	SEM+
	72590L1	Cover for SEM / SCL Silver	
	72590L2	Cover for SEM / SCL White	
	72590L0	Dummy Cover for Systevo Electronic Module White	
	72590T1	Namensschild Silber	
	72590T2	Namensschild Weiß	

7. Specifications

Systevo Bed Unit M	SEM+	SEM	SCL
APPROVALS and ENVIRONMENTAL			
Approval/Norms	CE, VDE 0834, UL 1069		
Environmental			
Operarting Temperature	+5°C .. +55°C		
Storage Temperature	-20°C .. +70°C		
Humidity	10% bis 90% (non-condensing)		
Environmental Sealing	IP 20 (installed with respective lamp cover /cover plate)		
Electrical			
Power Supply	24V DC +/- 10%		
Idle state means Current consumption (all LED off + nameplate connected)	60mA	52mA	?
Maximum state means Current consumption (all function working + nameplate LED&buzzer on)	140mA	130mA	?
DBUS IDs / Positions	3 identical devices on same DBUS line possible		
Passive input ports	0 V – 24 V +/- 10%		
Passive output ports	Max.100 mA open Collector to GND		
Wiring terminals- max. No of wires to be inserted	3 x 0,6mm / 2.x 0,8mm (non-flexible wires)		
Mechanical			
Dimensions of the Product (H x W x D)	?		
Installation depth	?		
Net weight	?		
Antimicrobial	Yes		
Fire protection class	V-0		
Housing / Enclosure material	PC + ABS-FR3010 (x% antimicrobial)		
Housing / Enclosure colour	White: Traffic white RAL 9016		